

- a) Find $T(u)$, the image of u under the transformation T .
 b) Find an x in $\mathbb{R}^2$ whose image under T is b .
 c) Is there more than one x whose image under T is b ?
 d) Determine if c is in the range of the transformation T .

Solution: a) $T(u) = Au = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2-3(-1) \\ 3 \cdot 2 + 5 \cdot (-1) \\ -2 + 7 \cdot (-1) \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$

b) We need to find out the vector x such that $T(x) = b$.
 $\therefore$ We need to solve the equation $Ax = b$.

By Theorem 3, the matrix equation $Ax = b$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ b]$.

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{bmatrix} \xrightarrow[\times(1/7)]{\text{Row 2 = Row 2} - 3 \times \text{Row 1}} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ -1 & 7 & -5 \end{bmatrix} \xrightarrow[\text{+ Row 1}]{\text{Row 3 = Row 3} + \text{Row 1}} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & -2 \end{bmatrix} \xrightarrow[\times 1/2]{\text{Row 2 = Row 2} \times 1/2} \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{bmatrix}$$

The corresponding system of equations are: ① $x_1 - 3x_2 = 3$ ② $x_2 = -1/2$ ③ $0 = 0$
 Substituting value of x_2 in ①, we get: $x_1 - 3(-1/2) = 3 \Rightarrow x_1 = 3/2$

$\therefore$ The vector x is: $\begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix}$

c) $\begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{+ } 3 \times \text{Row 2}]{\text{Row 1 = Row 1} + 3 \times \text{Row 2}} \begin{bmatrix} 1 & 0 & 3/2 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{bmatrix}$

Verification of whether the final matrix is in rref

- Row 1 and Row 2 are non-zero rows. Row 3 is a zero row. $\therefore$ The condition that all the non-zero rows must be above the zero rows is True.
- Row 1 Leading entry = 1 in column 1
 Row 2 Leading entry = 1 in column 2
 $\therefore$ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.